

Transaction Concept & ACID Properties -- DBMS

Definition: A Transaction is a logical unit of database processing that includes one or more database operations (READ, WRITE). To preserve data integrity, transactions must satisfy all four ACID Properties.

1. Detailed Technical Explanation

The ACID Properties:

A - ATOMICITY	: All operations complete successfully, or ALL are undone ("All-or-Nothing").
C - CONSISTENCY	: Transaction transforms database from one valid state to another valid state.
I - ISOLATION	: Concurrent transactions execute independently without interfering with each other.
D - DURABILITY	: Once committed, updates persist permanently in non-volatile storage even after system failure.

1. Atomicity

- * Enforced by the Recovery Manager using Write-Ahead Logging (WAL) and Rollback mechanisms.

2. Consistency

- * Enforced by database schema constraints (Primary Key, Foreign Key, CHECK constraints) and application logic.

3. Isolation

- * Enforced by the Concurrency Control Manager using locking or timestamping.

4. Durability

- * Enforced by Log Recovery & Checkpointing writing updates to non-volatile disk/SSD storage.

2. Transaction State Diagram



3. Core Concepts & Memory Keywords

- * ACID: Atomicity, Consistency, Isolation, Durability.

4. Must-Write Points for Exams

- * Atomicity guarantees that partial execution of a transaction is impossible.

5. Quick Recall Flow

Active -> Read/Write -> Partially Committed -> Commit Log Written -> Committed (Or Aborted on Failure)